

FRANK ALBANESE

CTO & Engineering Leader • AI/ML • Distributed Systems • Enterprise Platforms

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EXECUTIVE SUMMARY

CTO and Y Combinator founder (W21) with a track record of building technology companies from zero to exit. Over 10 years architecting AI-powered platforms, distributed systems, and data-driven applications across healthcare, enterprise SaaS, and Web3. Deep hands-on engineering expertise paired with strategic product vision—delivered mission-critical systems in regulated environments (HIPAA), scaled enterprise platform adoption, and driven applied AI/ML from research to production. Seeking a high-impact engineering role where I can apply the full breadth of my technical and leadership experience to hard problems at scale.

PROFESSIONAL EXPERIENCE

Co-Founder & CTO — CopyCatch.AI (Early-Stage)

Mar 2025 – Present

New York, NY

- Built the technical foundation for an AI-powered content authentication and intellectual property protection platform from concept to MVP.
- Architected scalable AI/ML pipelines for content analysis, pattern recognition, and automated detection at enterprise scale.
- Defined product engineering and technical strategy, aligning technology investments with go-to-market objectives.

Enterprise Customer Success Engineer — FlutterFlow (Y Combinator W21)

Jan 2024 – Sep 2025

New York, NY • Remote

- Served as senior technical partner for enterprise clients building mission-critical applications, ensuring scalable, secure deployments across industries.
- Architected custom solutions integrating AI/ML capabilities, cloud-native services (Firebase, GCP, AWS), and third-party APIs for complex enterprise requirements.
- Drove product strategy feedback loops between enterprise customers and engineering leadership, directly influencing platform roadmap and feature prioritization.
- Led technical enablement initiatives, developing best practices for security, performance optimization, and distributed application architecture.

Co-Founder & Chief Technology Officer — CoVerified — COVID-19 Campus Management Platform

May 2020 – Jun 2021

New York, NY • Y Combinator W21

- Defined and executed the full technology vision for a Y Combinator-backed (W21) venture, building a mission-critical campus health management platform from concept through production and successful exit.
- Designed and architected secure, scalable distributed systems handling sensitive health data with compliance-first engineering practices.
- Recruited, mentored, and led a cross-functional engineering team across front-end, back-end, and data engineering disciplines.
- Drove applied AI/ML integration for predictive modeling, risk assessment, and data-driven decision-making at scale.
- Partnered with CEO and stakeholders to align technology investments with strategic objectives and fundraising milestones.

Software Engineer & Data Analyst — Henry Ford Health System

Feb 2016 – Apr 2020

Greater Detroit Area

- Engineered data-driven web applications over 4+ years in collaboration with the Department of Emergency Medicine, delivering real-time analytics and predictive tools for clinical decision-making.

- Designed and built scalable data pipelines and visualization platforms processing large-scale healthcare datasets for operational insights.
- Applied machine learning techniques for predictive modeling and pattern recognition in healthcare data, supporting evidence-based clinical strategies.
- Ensured HIPAA-compliant data handling, security protocols, and audit-ready documentation across all systems.

TECHNICAL COMPETENCIES

AI & Machine Learning: Generative AI, Computer Vision, Predictive Modeling, NLP, Content Analysis, Applied ML

Languages & Frameworks: Python, TypeScript, JavaScript, Dart/Flutter, Solidity, Node.js, React

Data Engineering: SQL/NoSQL, Data Pipelines, ETL, Real-Time Analytics, Large-Scale Data Processing

Cloud & Infrastructure: GCP, AWS, Firebase, Docker, Kubernetes, CI/CD, Infrastructure-as-Code

Distributed Systems: Microservices, Event-Driven Architecture, Blockchain/Web3, Decentralized Systems

Security & Compliance: HIPAA, Cybersecurity, Secure SDLC, Compliance Frameworks, Data Privacy

EDUCATION

Y Combinator — W21 Batch

Founder • YC Founder Verification • 2021

Oakland University

Bachelor's Degree, Management Information Systems • 2013 – 2015

Activities: TEDx Oakland University, Cyber Security OU

AWARDS

- **2023 NextGrid Ventures + LabLab AI Startup Hackathon — Winner** (lablab.ai, Feb 2023)

SELECT PROJECTS & VENTURES

- **AI-Powered Art Generation:** Implemented and extended Nvidia's StyleGAN2 for generative AI portrait creation, demonstrating applied deep learning and computer vision expertise.
- **Algorithmic Trading Systems:** Designed and built autonomous trading algorithms and bot infrastructure for real-time financial market execution (Solana, traditional markets).
- **Tavern Community Coworking (Co-Founder):** Architected the technology platform for a community-driven coworking network, including member management, space optimization, IoT integration, and operational automation.
- **Blockchain Governance:** Active contributor to decentralized governance frameworks (Cardano Catalyst via Snapbrillia), advancing secure, community-driven decision-making at scale.

COMMUNITY

- **Volunteer:** Toys For Tots